

Morphological image processing

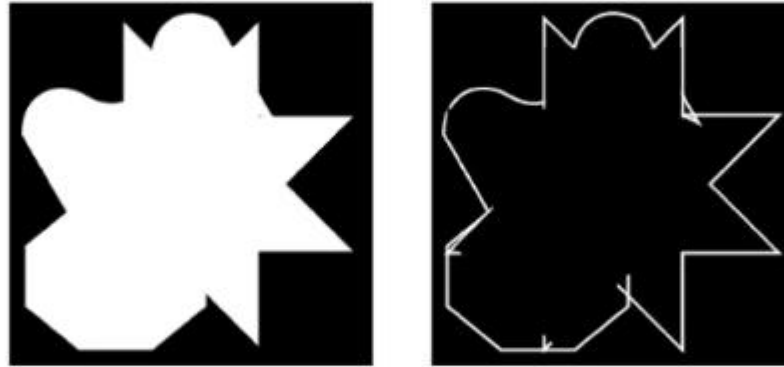
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Overview

- *morphology* is a tool for extracting image components
- useful in the description of region shape, such as boundaries, skeletons, and convex hulls
- mathematical morphology is set theory, it can apply directly to binary (two-level) images
 - either in the set (a pixel is set, or put to foreground)
 - or it isn't (a pixel is reset, or put to background)
- the usual set operators (intersection, union, inclusion, complement) can be applied to them

Overview continues

- Basic operations in mathematical morphology operate on two sets:
 - the first one is the *image*, and
 - the second one is the *structuring element*
- The structuring element used in practice is generally much smaller than the image, often a 3x3 matrix
- Applied on greyscale images too



Boundary extraction using morphological image processing operations

Types of morphological operations

- The structuring element's size and shape help us determine how many pixels are added to or subtracted from the object in the image. We define the morphological operation as **dilatation** or **erosion** by the rule that processes the pixels.
- **Dilation**: In the dilation process, we add pixels to the bounded objects.
- **Erosion**: In the erosion process, we remove pixels from the edges of objects.

EROSION

- The basic effect of erosion operator on a binary image is to erode away the boundaries of foreground pixels (usually the white pixels).
- Thus areas of foreground pixels shrink in size, and "holes" within those areas become larger.
- Mathematically, erosion of set A by set B is a set of all points x such that B translated by x is still contained in A .

EROSION continues

- Let foreground pixels be represented by logical 1's, and background pixels by logical 0's.
- As a practical example, take a 3x3 matrix of logical 1's, with the middle point chosen as the origin of the set is used as the structuring element B (see the following picture).
- To compute the erosion of a binary input image by this structuring element, consider each of the foreground pixels in the input image in turn.

EROSION EXAMPLE



DILATION

- The basic effect of dilation on binary images is to enlarge the areas of foreground pixels (*i.e.* white pixels) at their borders.
- The areas of foreground pixels thus grow in size, while the background "holes" within them shrink.
- Mathematically, erosion of A by B is the set of all x displacement such that B and A overlap by at least one non-zero element.

DILATION continues

- Again let us take a 3x3 matrix for the structuring element, with the center pixel used as the origin of the set. B , then the dilation can be performed using the logical OR function:
 - If the pixel is set to foreground, it remains such.
 - If the pixel is set to background, but at least one of its eight neighbours is set to foreground, the pixel is converted to foreground.
 - If the pixel is set to background and none of its eight neighbours is set to foreground, the pixel remains set to background.

DILATION EXAMPLE



OPENING AND CLOSING

- The **opening** is a composite operator, constructed from the two basic operators described above.
- Opening of set A by set B is achieved by first the eroding set A by B , then dilating the resulting set by B .

OPENING EXAMPLE

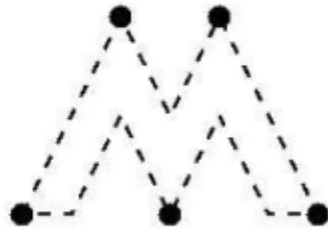


CLOSING

- The **closing**, like opening, is also a composite operator.
- The closing of set A by set B is achieved by first dilating of set A by B , then eroding the resulting set by B .

Hit-and-Miss Transform

- The hit-and-miss transform is a general binary morphological operation that can be used to look for particular patterns of foreground and background pixels in an image.



Common Names: Hit-and-miss Transform, Hit-or-miss Transform

Hit-and-Miss Transform

- It is actually the basic operation of binary morphology since almost all the other binary morphological operators can be derived from it.
- As with other binary morphological operators it takes as input a binary image and a structuring element, and produces another binary image as output.

Morphological Gradient

The morphological gradient is the difference between the dilation and erosion of an image.

- **Purpose:** To highlight the boundaries or edges of objects in the image.
- **How it Works:** By subtracting the eroded image from the dilated image, the boundaries of the objects are emphasized.

Top-Hat Transform

The top-hat transform is used to extract small elements and details from an image.

- **Purpose:** To enhance bright objects on a dark background or vice versa.
- **How it Works:** There are two types: white top-hat (original image minus the result of opening) and black top-hat (result of closing minus the original image).

Original Image



Erosion



Dilation



Opening



Closing



Output of the Morphological Operations for Image Processing